

① Electrostatic 'Potential Energy':

for more than 2 charge

$$U = \sum \text{PE of all pair of charges}$$

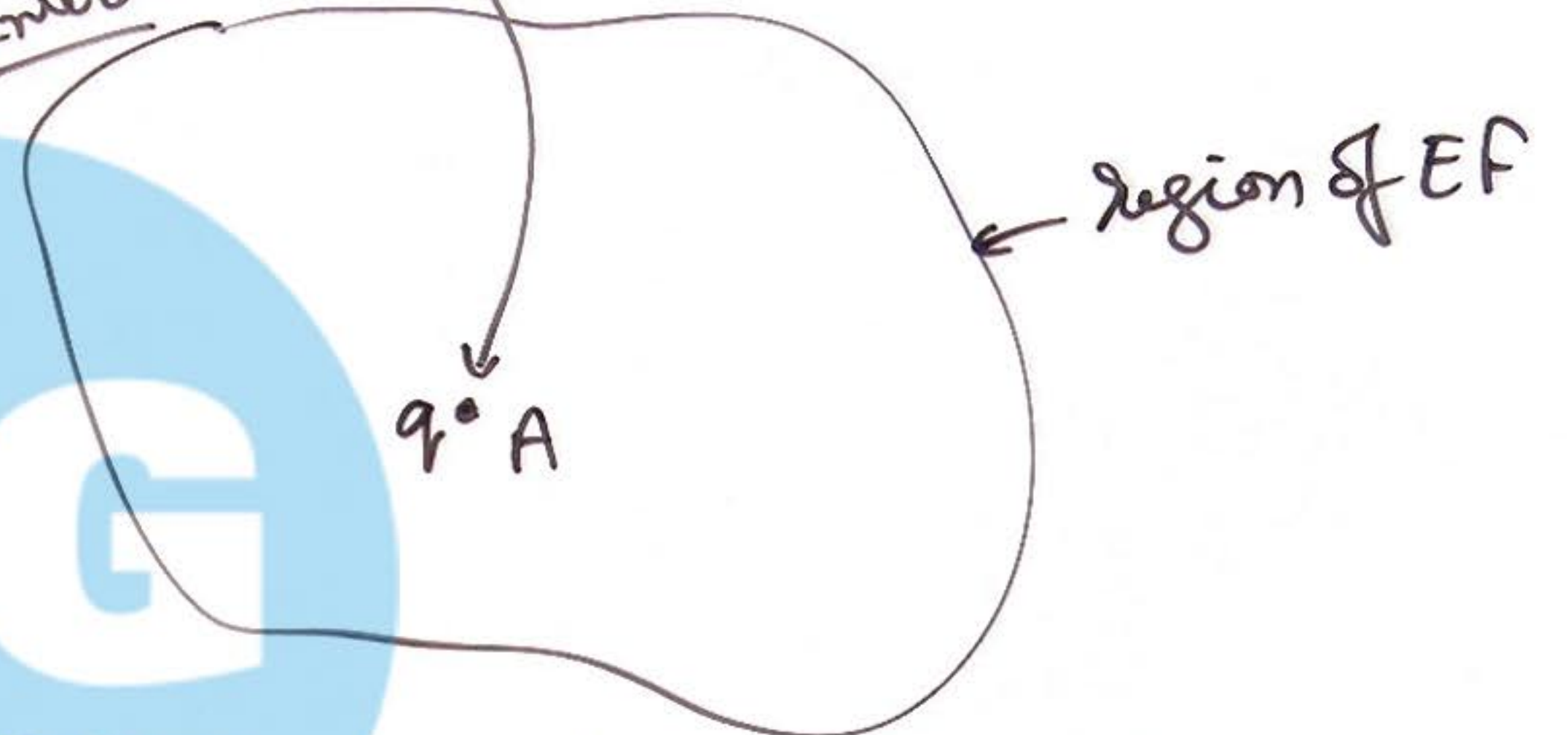
PE of two charge system



$$U = \frac{kq_1q_2}{r}$$

work done in assembling a system.

$U=0$
State of No Interaction

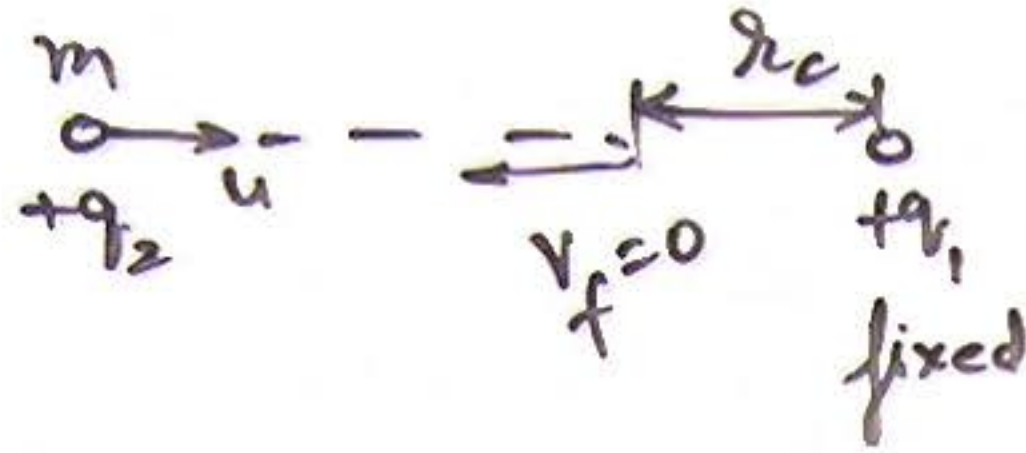


$U_{qA} = \text{W/D in shifting } q \text{ from } \infty \text{ to } A$

$$U_{qA} = - \int_{\infty}^A q \vec{E} \cdot d\vec{r}$$

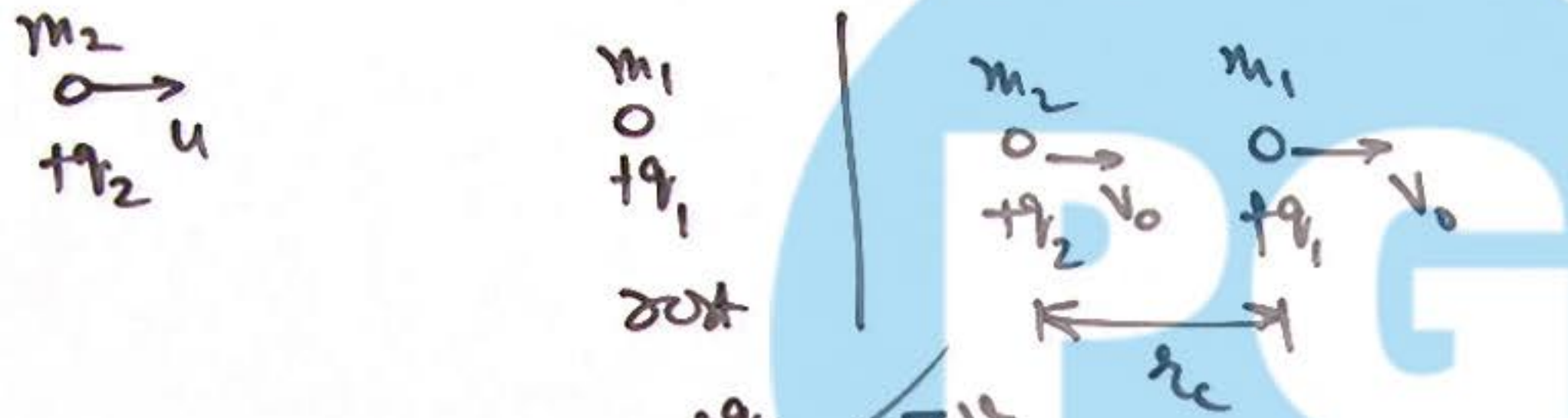
② Closest Distance of Approach:

Case-1



$$\frac{1}{2} m u^2 = \frac{k q_1 q_2}{r_c} \Rightarrow r_c = \frac{2 k q_1 q_2}{m u^2}$$

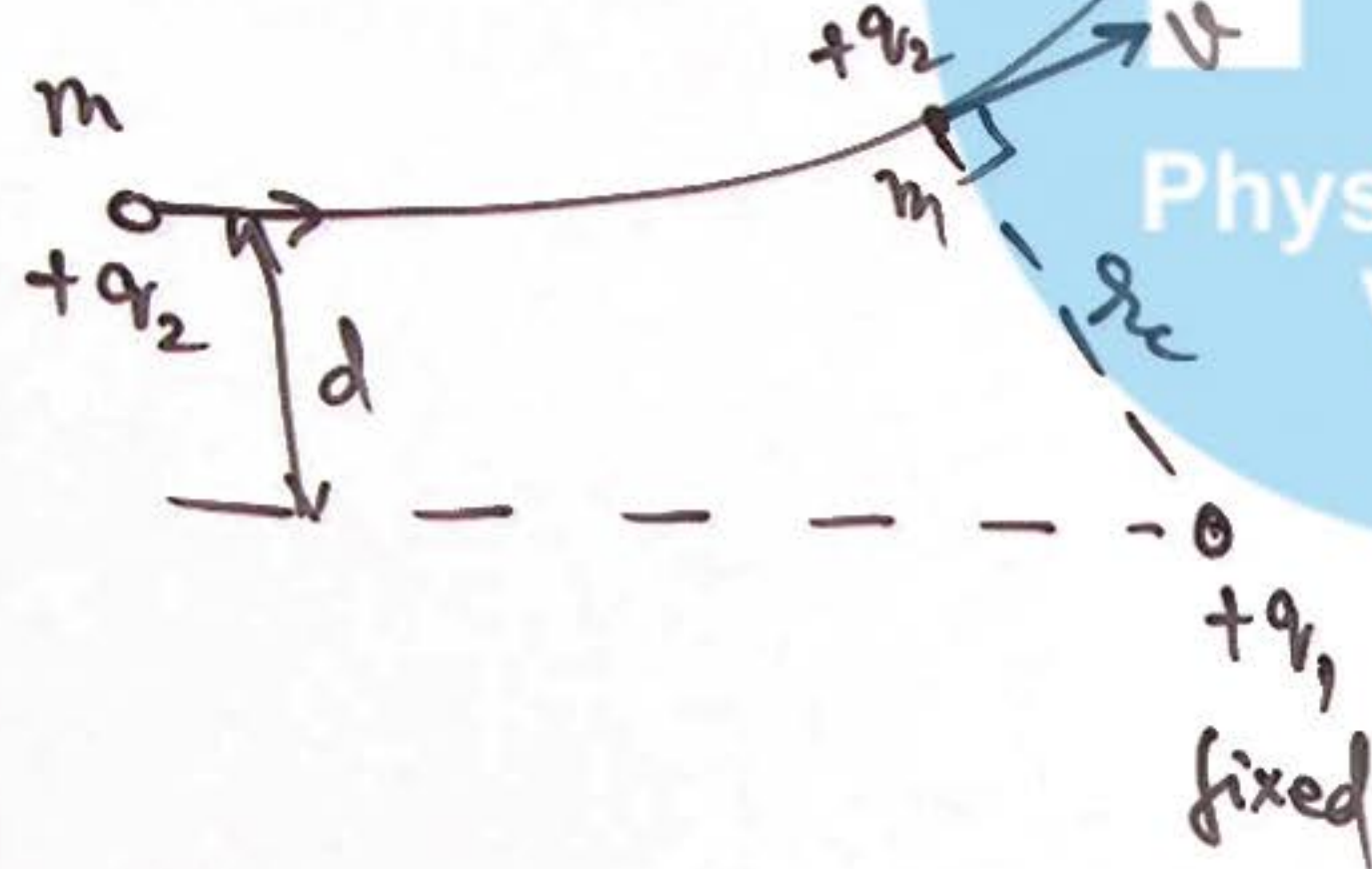
Case-2



$$\frac{1}{2} m u^2 = \frac{1}{2} (m_1 + m_2) v_0^2 + \frac{k q_1 q_2}{r_c}$$

$$\cancel{m_1} u + m_2 v = (m_1 + m_2) v_0$$

Case-3



$$\begin{cases} \frac{1}{2} m u^2 = \frac{1}{2} m v^2 + \frac{k q_1 q_2}{r_c} \\ m u d = m v r_c \end{cases} \Rightarrow r_c = \frac{2 k q_1 q_2}{m u^2} + \frac{d^2}{u^2}$$

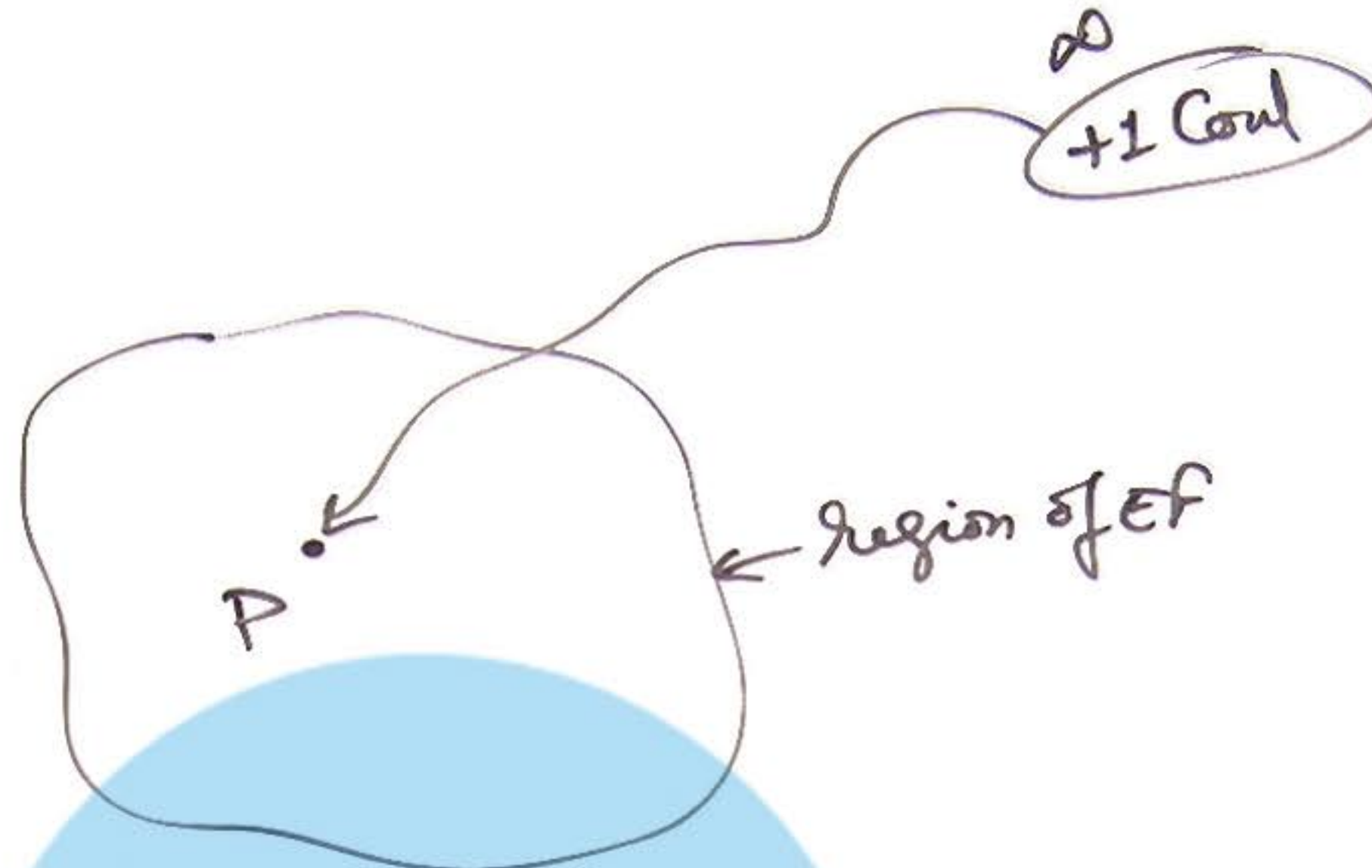
③ Electric Potential :

PE of a unit
pos charge at
a pt in space.

Scalar.



$$V_C = \frac{kq_1}{r_1} + \frac{kq_2}{r_2} + \frac{kq_3}{r_3}$$

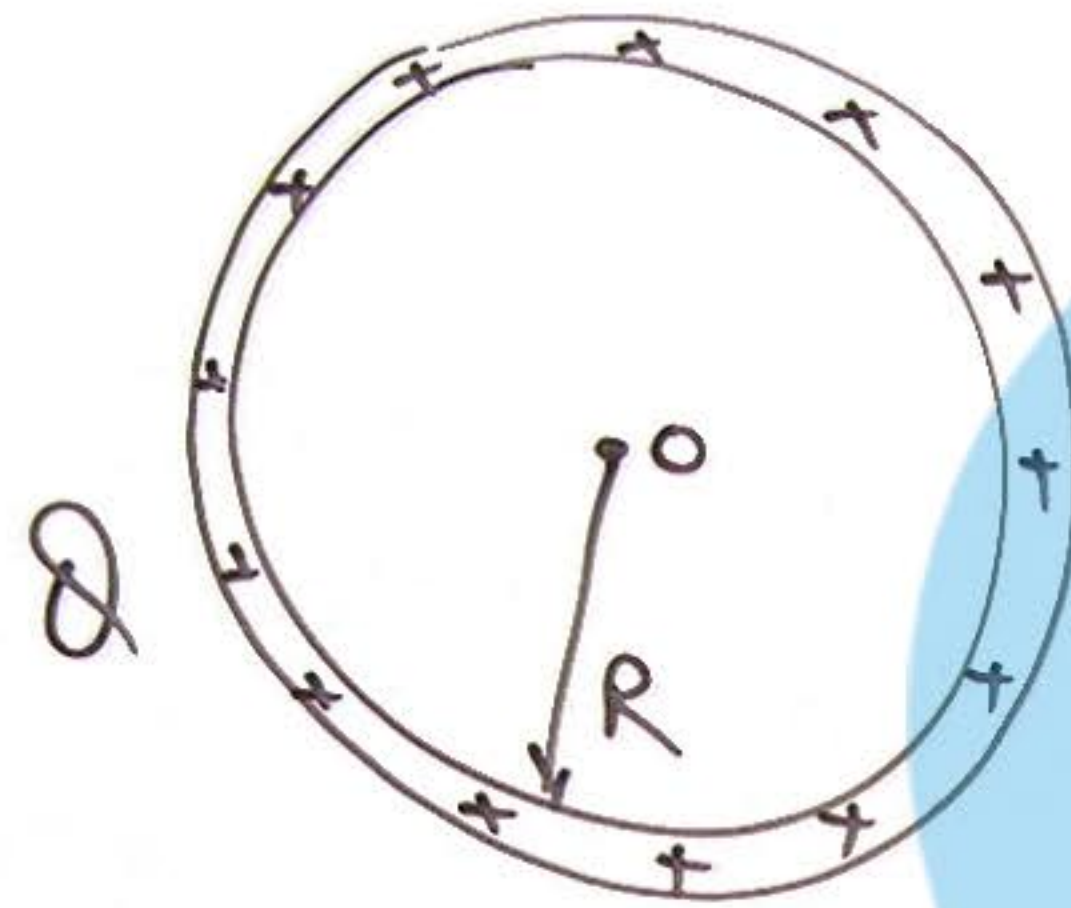


$$V_P = - \int_{\infty}^P \vec{E} \cdot d\vec{r}$$

[volt] / [J/C]

A diagram showing a point charge $+q$ at a distance r from a point P. A box contains the formula $V_P = \frac{kq}{r}$.

④ EP due to a charged Ring :



$$V_0 = \frac{kQ}{R}$$

for uniformly or
non uniformly
Charge ring

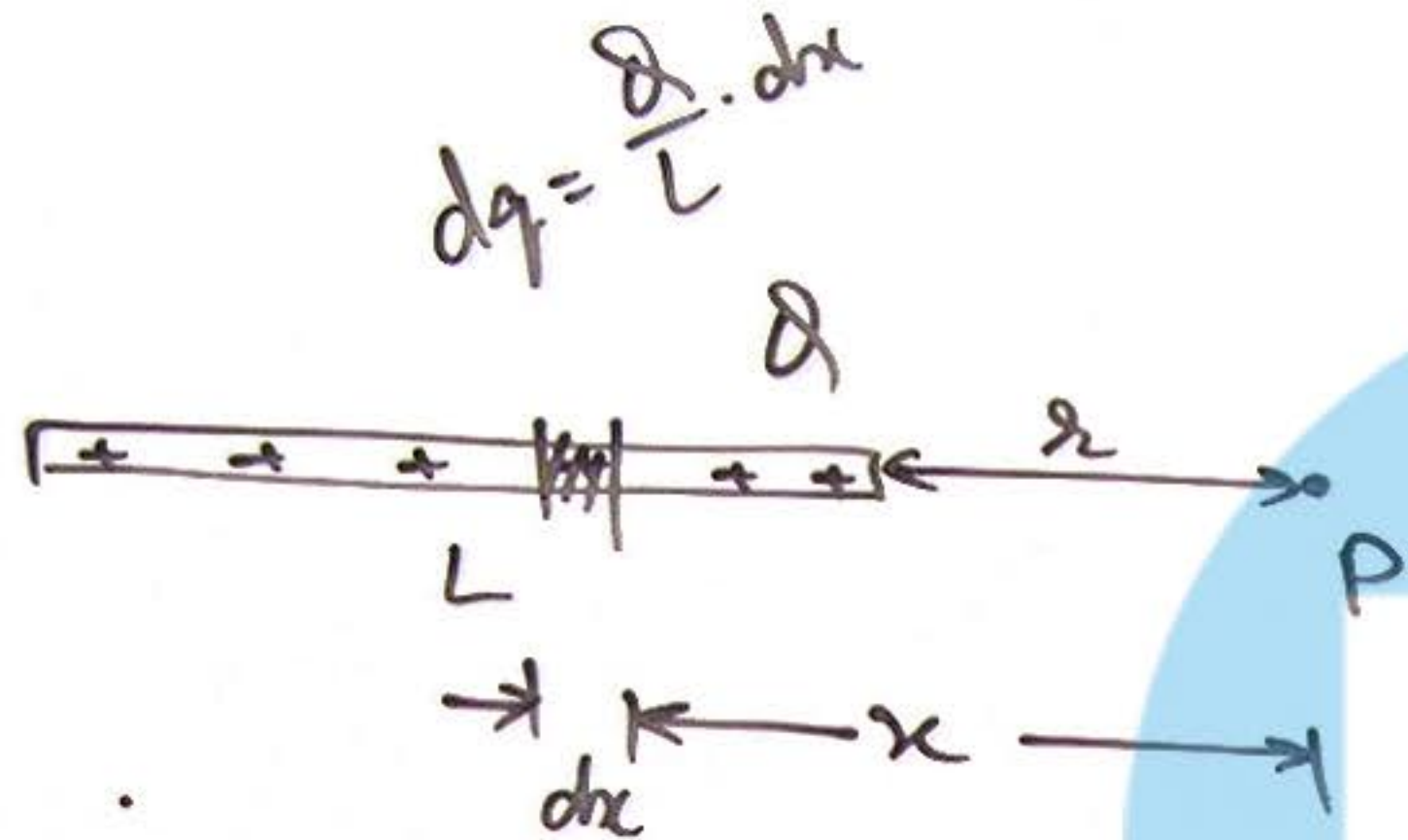


$$V_P = \frac{kQ}{\sqrt{x^2 + R^2}}$$

$$V_P = - \int_{\infty}^x \vec{E} \cdot d\vec{r}$$

$$\frac{kQx}{(x^2 + R^2)^{3/2}}$$

⑤ EP due to a charged Rod :

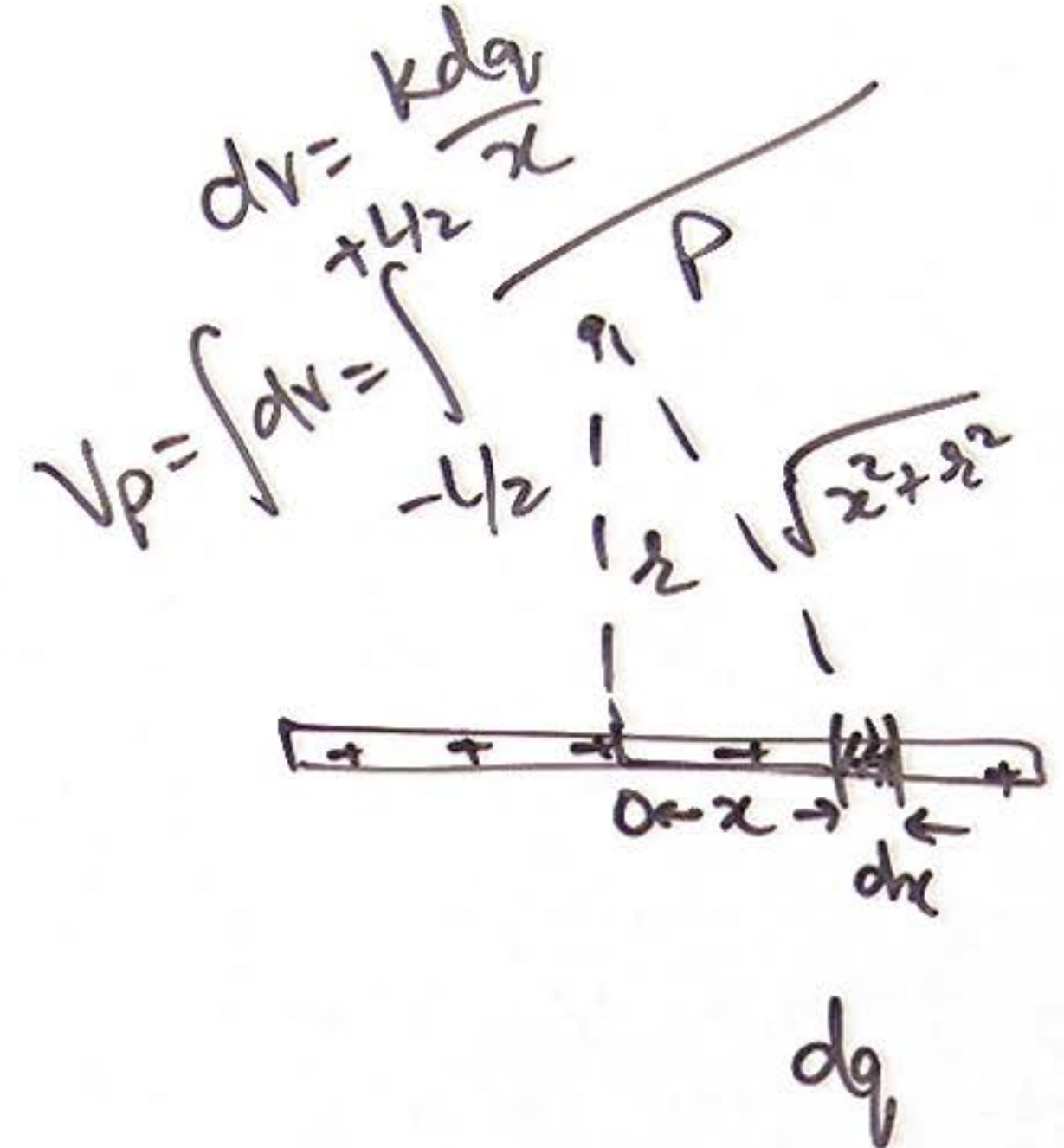


$V_P \rightarrow$ Not defined

$$dV = \frac{k dq}{x}$$

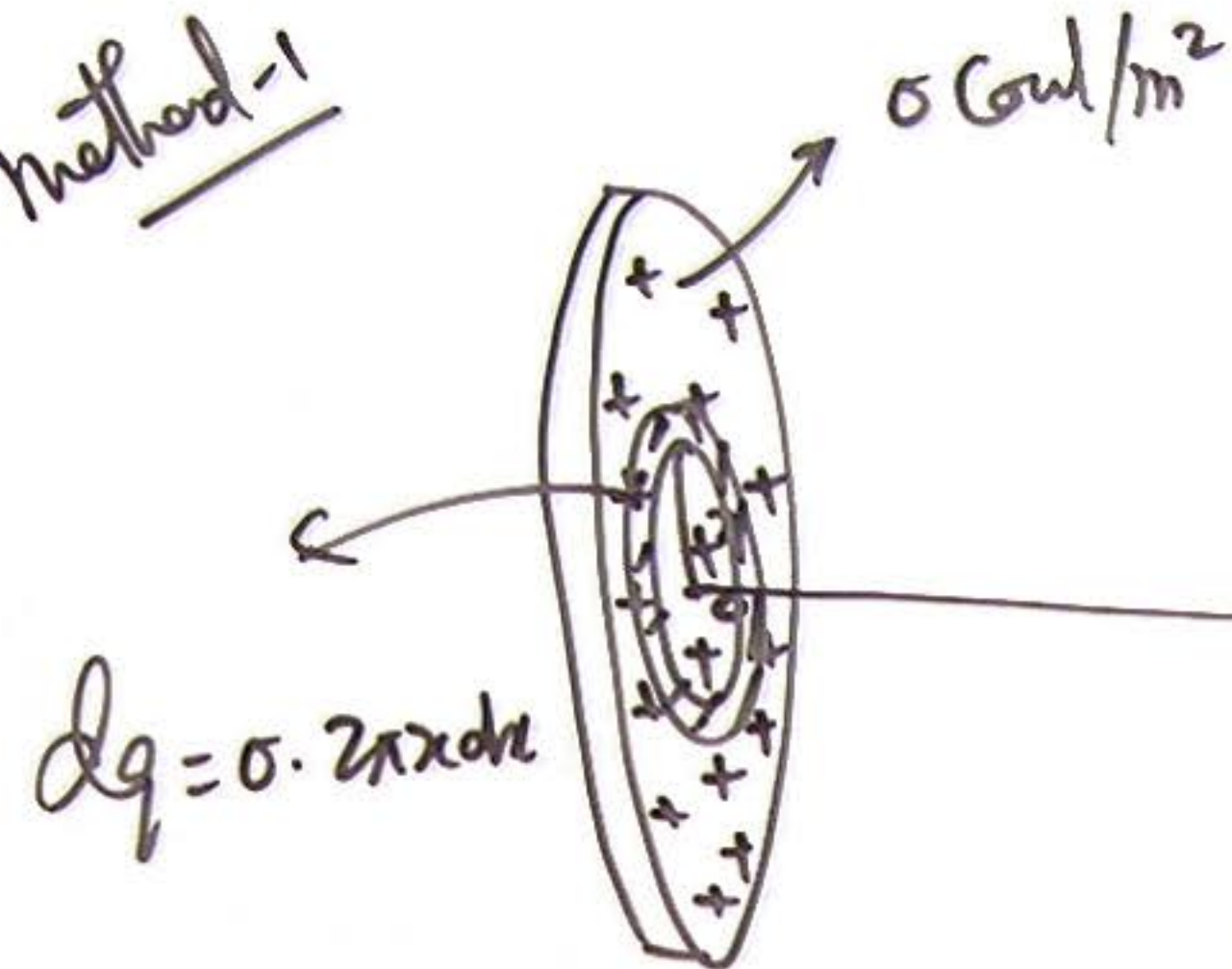
$$V_P = \int dV = \int \frac{k Q}{L} \cdot \frac{1}{x} dx$$

$$V_P = \frac{kQ}{L} \ln\left(\frac{r+L}{r}\right)$$



⑥ EP due to a uniformly charged disc:

Method-1



$$dV = \frac{k dq}{x} = \int_0^R \frac{k \sigma 2\pi r dr}{x} = - \int_0^R \frac{k \sigma 2\pi r dr}{x}$$

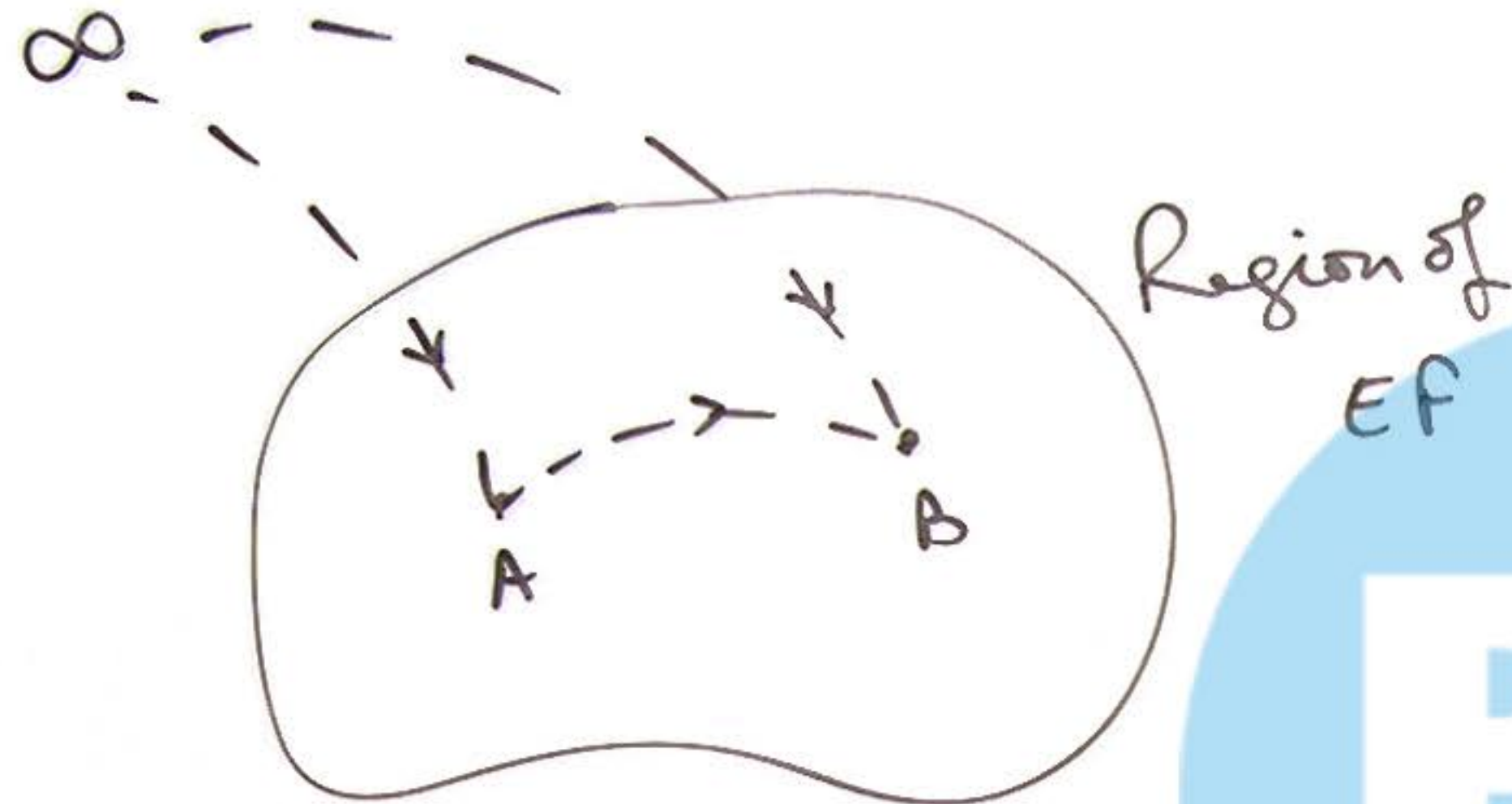
$$V_0 = \frac{\sigma R}{2\epsilon_0}$$

$$E = \frac{\sigma}{2\epsilon_0} \left[1 - \frac{x}{\sqrt{x^2 + R^2}} \right]$$

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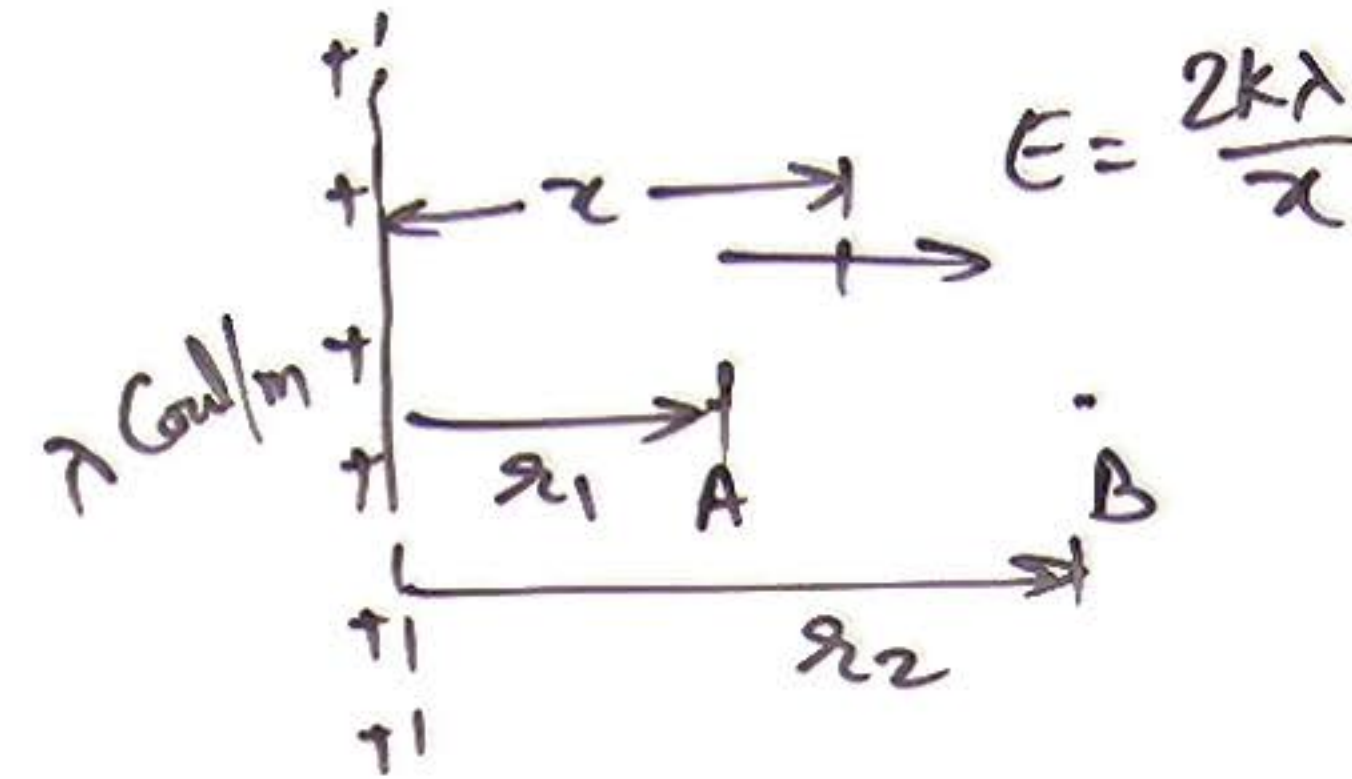
$$V_0 = - \int_{\infty}^0 \vec{E} \cdot d\vec{r} = ()$$

⑦ Calculation of Potential Difference:



$$V_A + \left(- \int_A^B \vec{E} \cdot d\vec{r} \right) = V_B$$

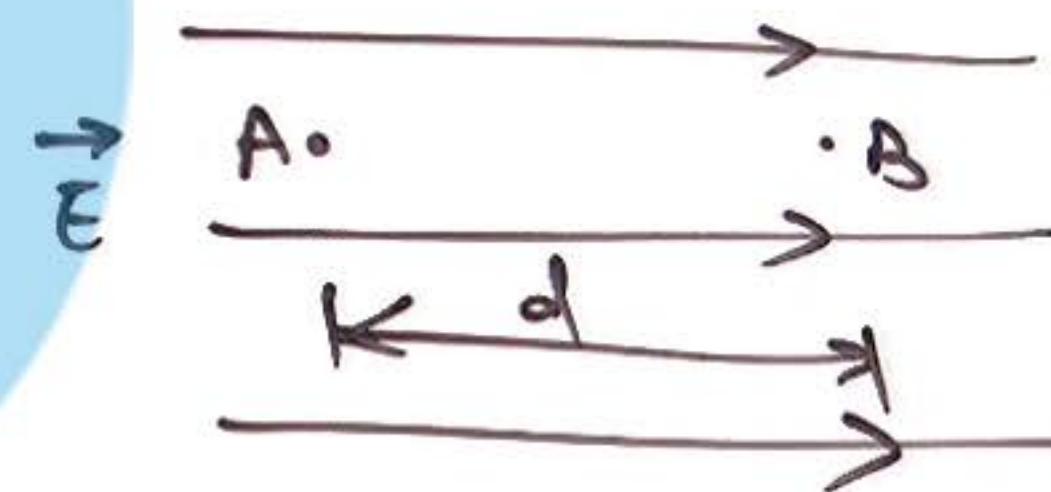
$$V_B - V_A = - \int_A^B \vec{E} \cdot d\vec{r}$$



$$V_A - V_B = \left| \int_{r_1}^{r_2} \frac{2k\lambda}{r} dr \right| = 2k\lambda \ln\left(\frac{r_2}{r_1}\right)$$

NOTE: In dir of EF, Electric Potential decreases

Uniform EF

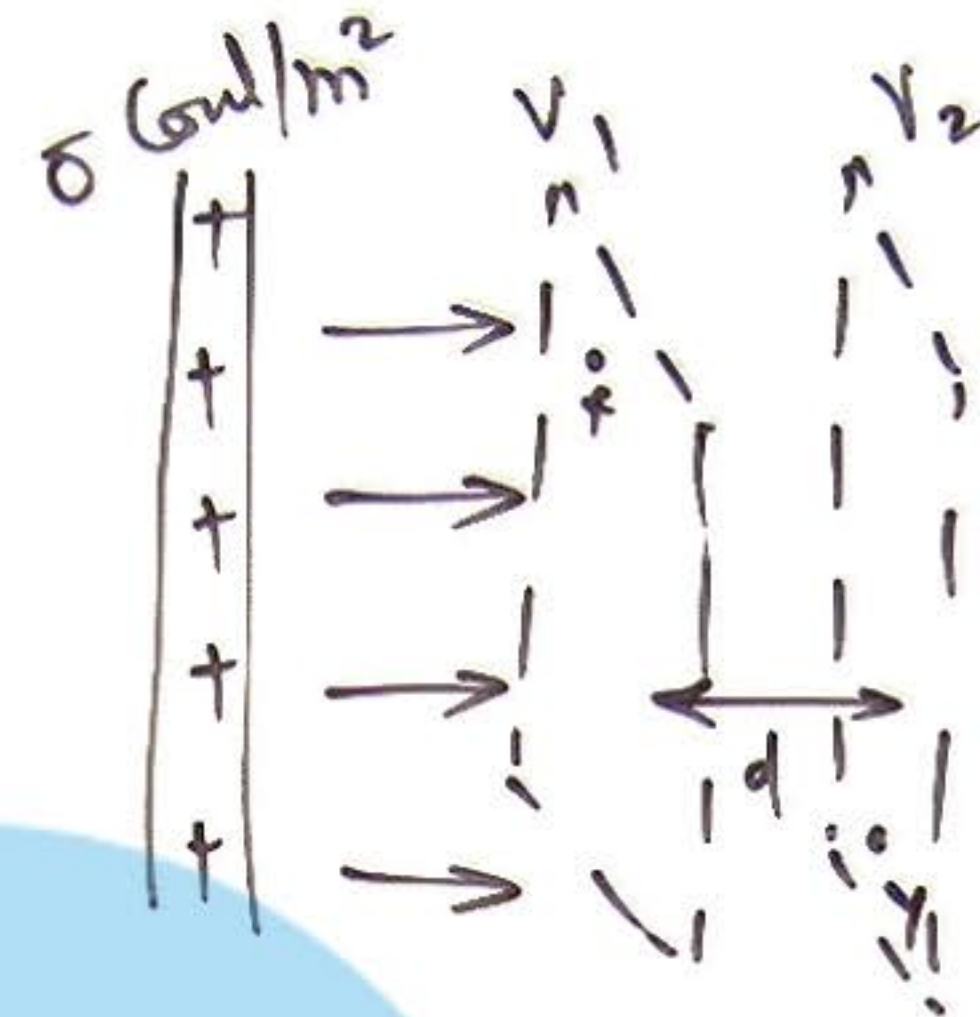


$$V_A - V_B = Ed$$

⑧ Equipotential Surfaces:

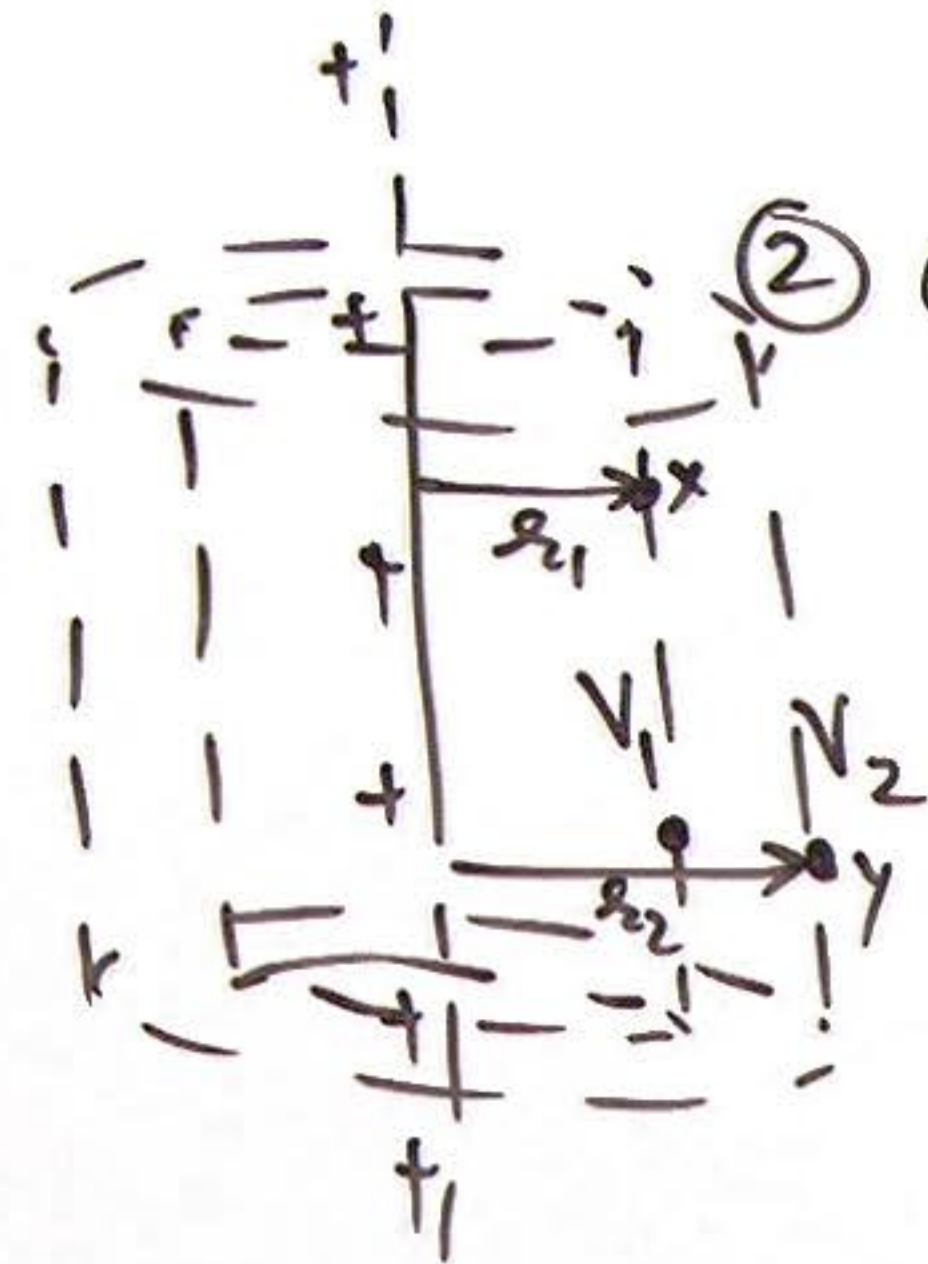
Condition

① EF is $\perp$ to equipotential surface at all pts

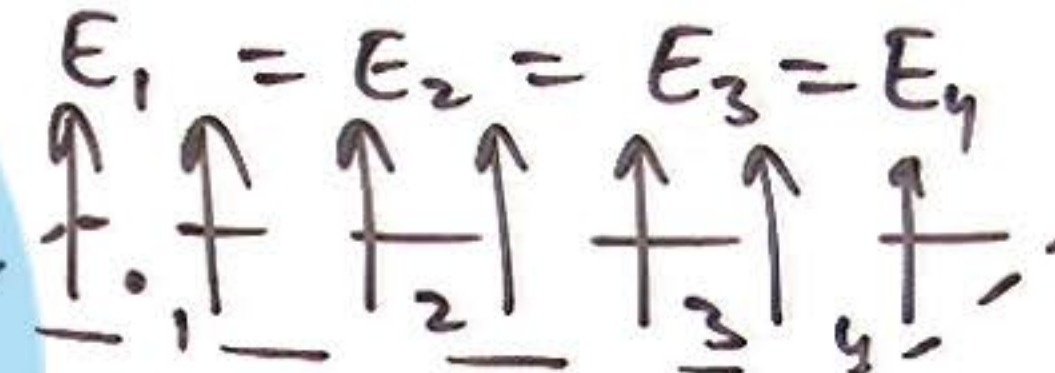


$$E = \frac{\sigma}{2\epsilon_0}$$

$$V_1 - V_2 = Ed = \frac{\sigma d}{2\epsilon_0}$$



② EF is either constant or zero at all pts of surface



$$E_1 = E_2 = E_3 = E_4$$

$$V_1 > V_2 > V_3$$

$$V_1 > V_2 > V_3$$

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Concentric Spheres

⑨ Relation in EF & EP:

$$V_P = - \int_{\infty}^P \vec{E} \cdot d\vec{r}$$

EF in a region is gradient
of Potential in space.

$$\vec{E} = -\nabla V$$

$$[As \vec{F} = -\nabla U]$$

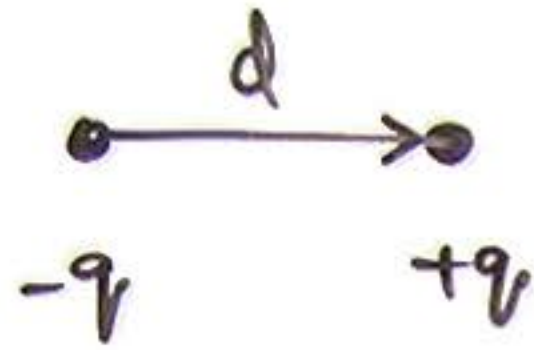
$$\vec{E} = - \left[\frac{\partial V}{\partial x} \hat{i} + \frac{\partial V}{\partial y} \hat{j} + \frac{\partial V}{\partial z} \hat{k} \right]$$

for unidirectional variation in V

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$$\vec{E} = - \frac{dV}{dr} \hat{r}$$

⑩ Electric Dipole :



Potential at P due to dipole

$$V = \frac{kP \cos \theta}{r^2}$$

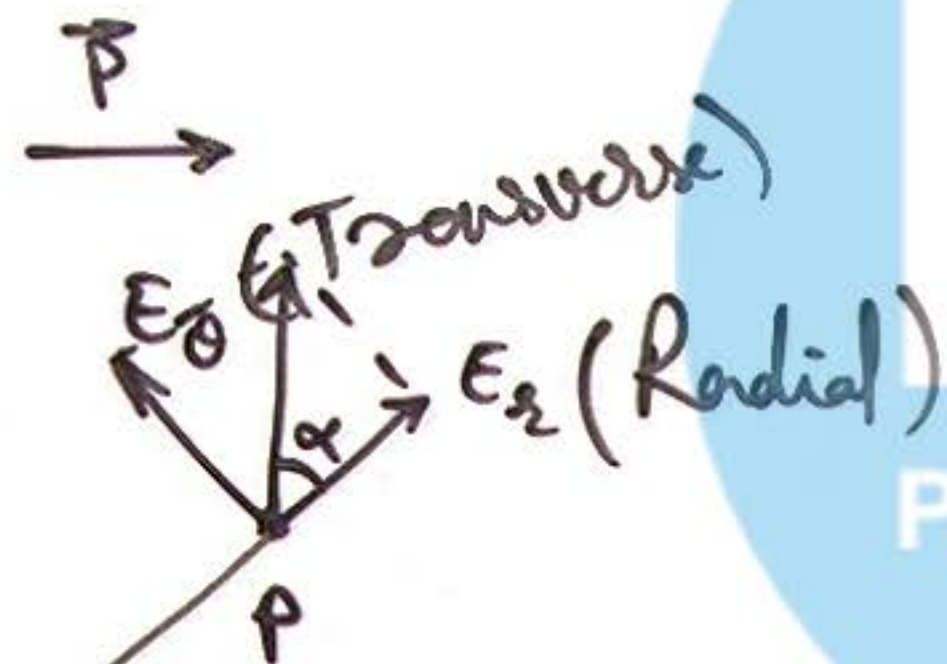
Dipole moment $\vec{P} = q\vec{d}$

$E_{eq} = \frac{kP}{r^3}$

EF due to a dipole

Equator

Axis of Dipole



$$E_r = \frac{2kP \cos \theta}{r^3}$$

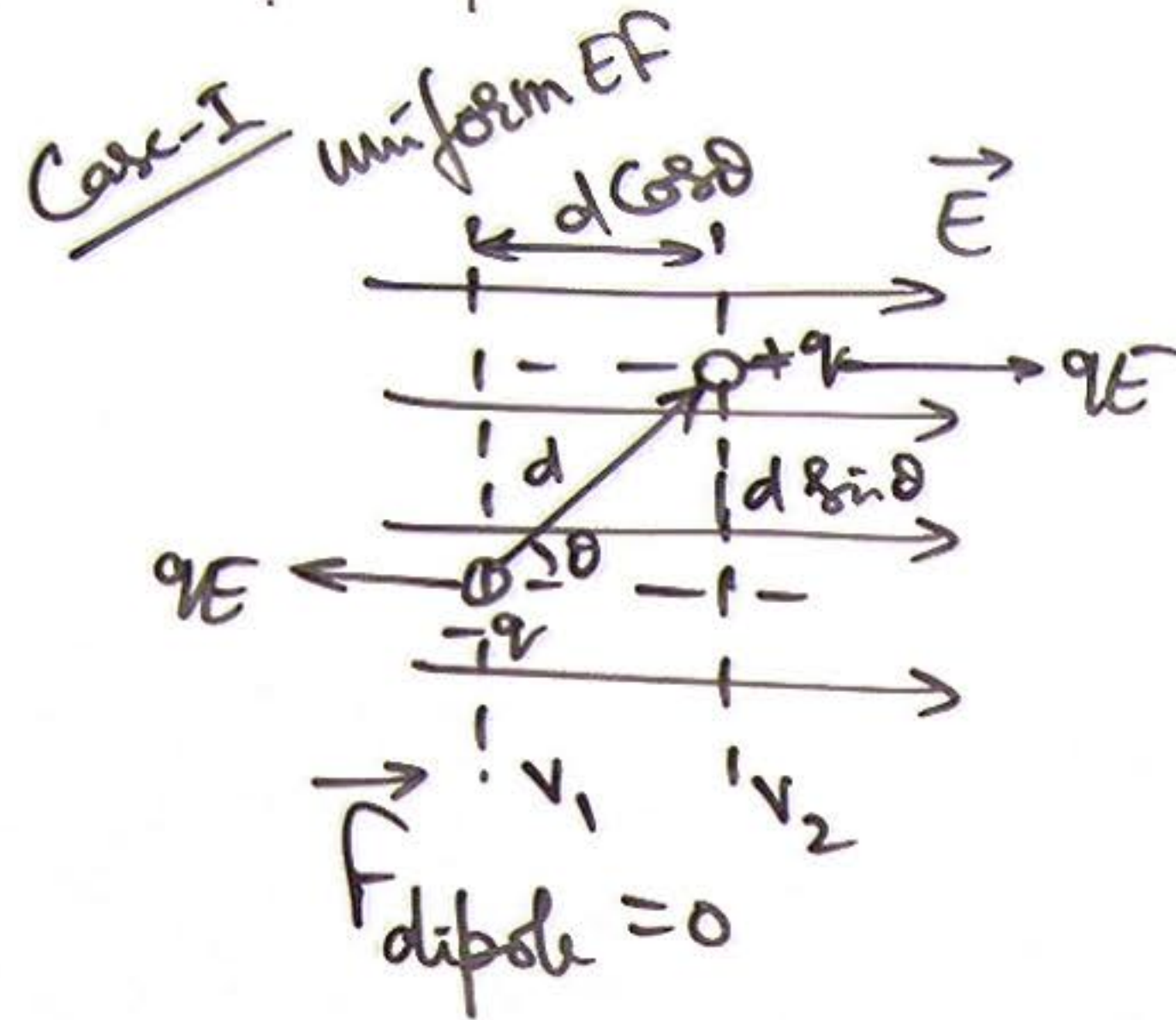
$$E_\theta = \frac{kP \sin \theta}{r^3}$$

$$\alpha = \tan^{-1} \left(\frac{1}{2} \tan \theta \right)$$

$$E_{axis} = \frac{2kP}{r^3}$$

$$E_P = \sqrt{E_\theta^2 + E_r^2} = \frac{kP}{r^3} \sqrt{1 + 3 \cos^2 \theta}$$

⑪ Dipole placed in EF:



$$\tau_{\text{couple}} = qE \times d \sin \theta = PE \sin \theta$$

Equal & oppo forces
not along same line

$$\vec{\tau} = \vec{P} \times \vec{E}$$

PE of dipole in EF $U = qV_2 - qV_1$

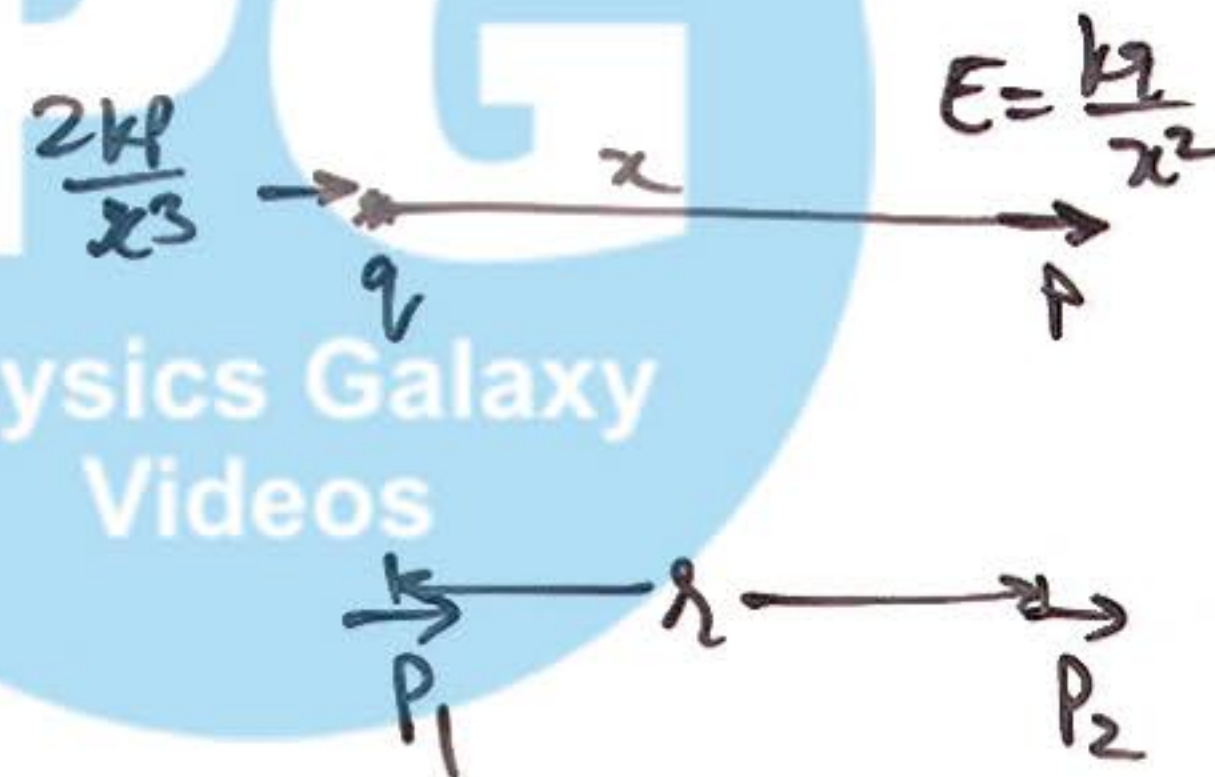
$$V_1 - V_2 = E(d \cos \theta)$$

$$U = -qEd \cos \theta$$

$$U = -PE \cos \theta$$

$$U = -\vec{P} \cdot \vec{E}$$

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Case-II in Non uniform EF:

PE $U = -\vec{P} \cdot \vec{E}$

force on dipole in NU EF is

$$\vec{F} = -\frac{dU}{dr} \hat{r}$$

$$\vec{F} = \left(\vec{P} \cdot \frac{d\vec{E}}{dr} \right) \hat{r}$$

$$\vec{F} = \frac{d}{dr} (\vec{P} \cdot \vec{E}) \hat{r}$$

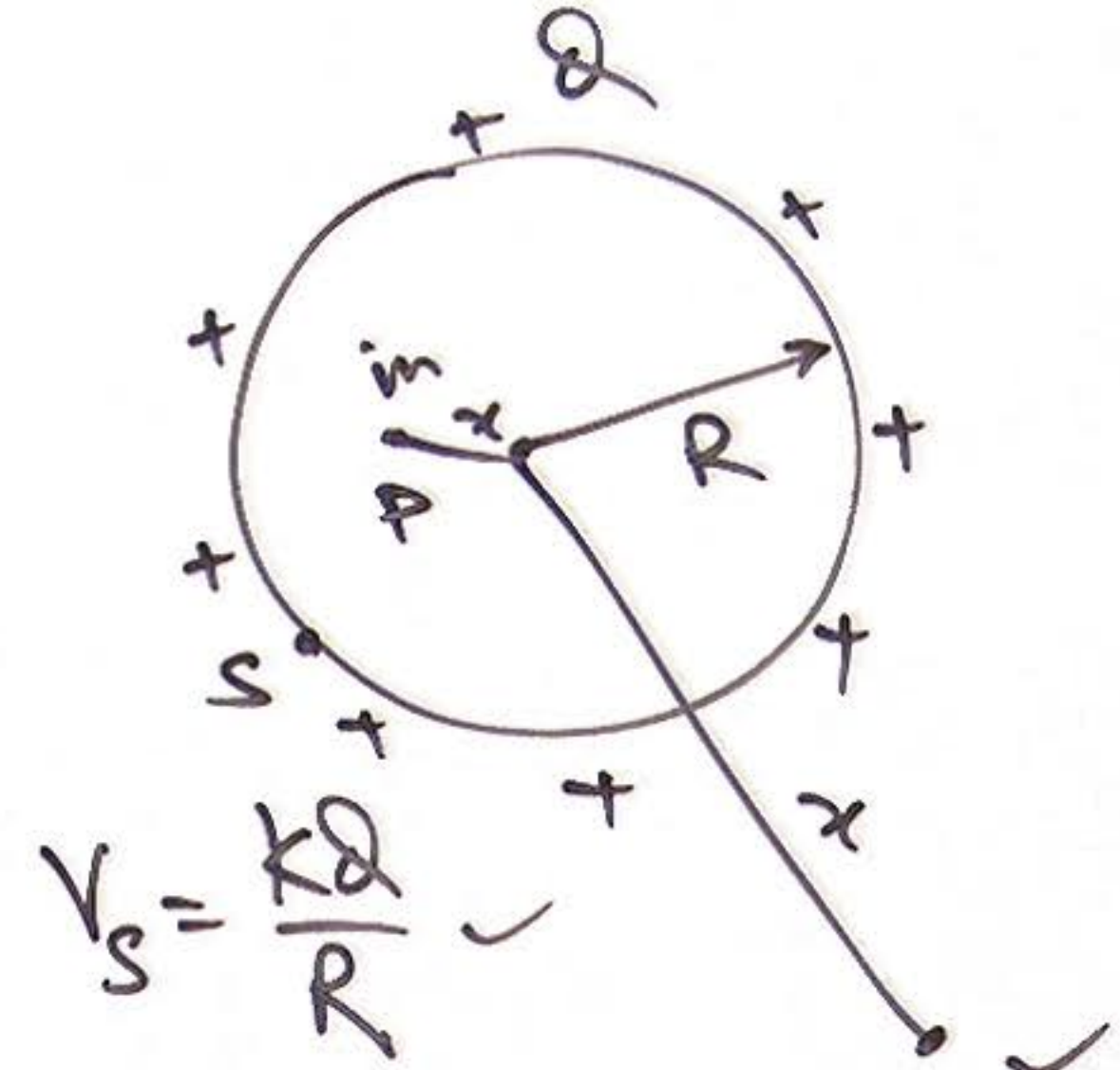
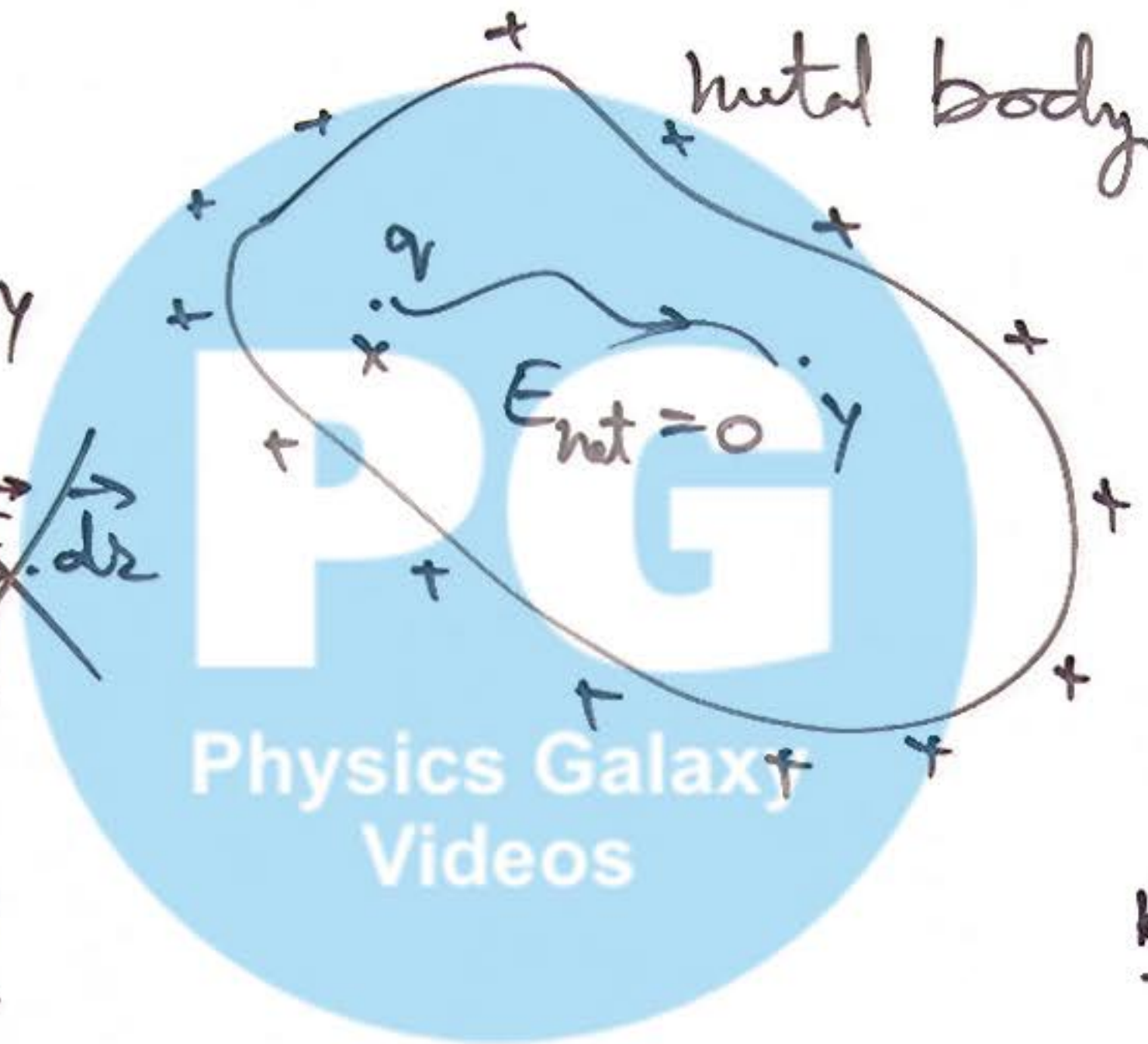
12) Potential inside a Conducting Body:

remain constant

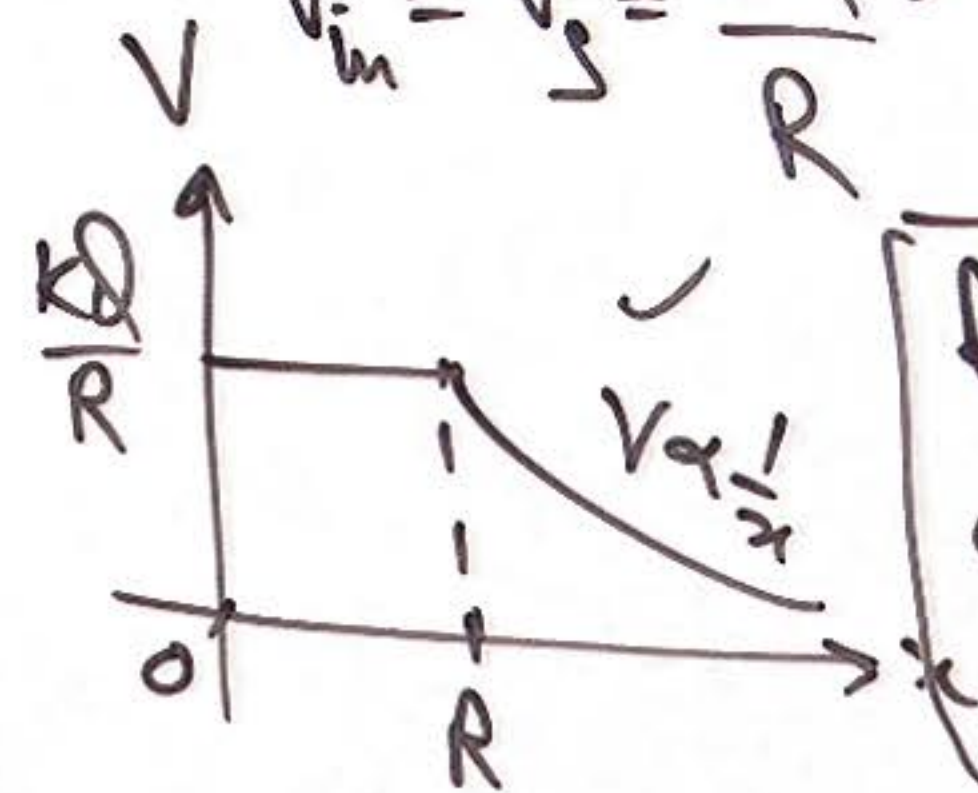
Potential diff betⁿ x & y

$$V_x - V_y = - \int_x^y \vec{E} \cdot d\vec{s}$$

$$V_x = V_y$$



$$V_{in} = V_s = \frac{KQ}{R} \quad V_{out} = \frac{KQ}{x}$$



Also valid for
a hollow unif ch
sphere

(13) Potential due to a Non-Conducting
Uniformly Charged Sphere:

$$V_{in} = \frac{kQ}{2R^3} (3R^2 - x^2)$$

at $x=0$

$$V_c = \frac{3}{2} \frac{kQ}{R} = \frac{3}{2} V_s$$



$$V_{out} = \frac{kQ}{x}$$

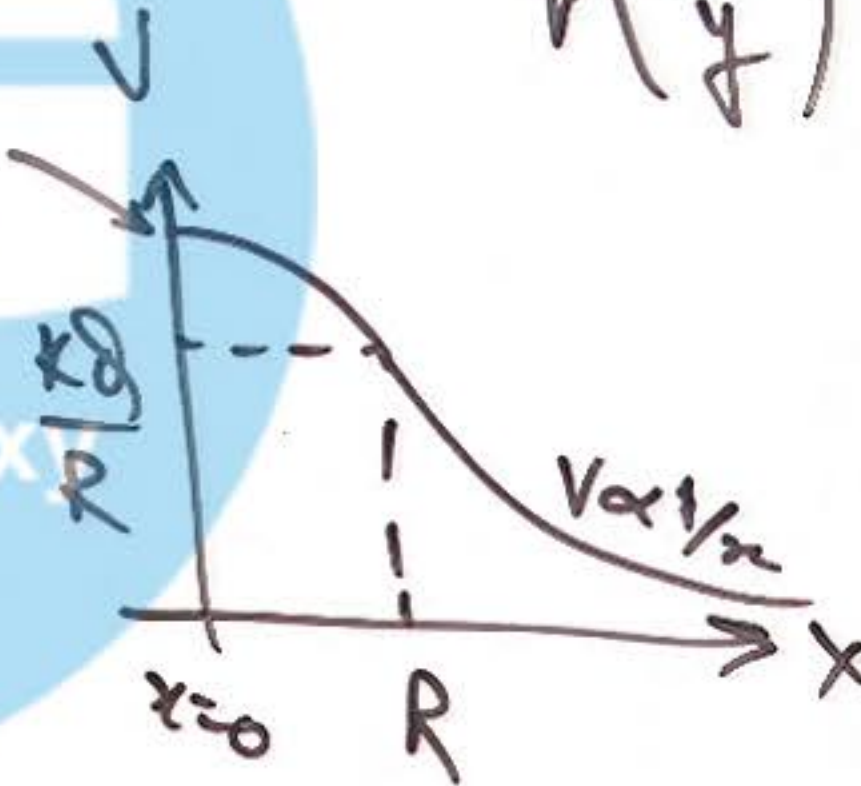
$$V_s = \frac{kQ}{R}$$

Q: Find bullet speed v so that it can reach pt M.

Sol: by Energy Cons.

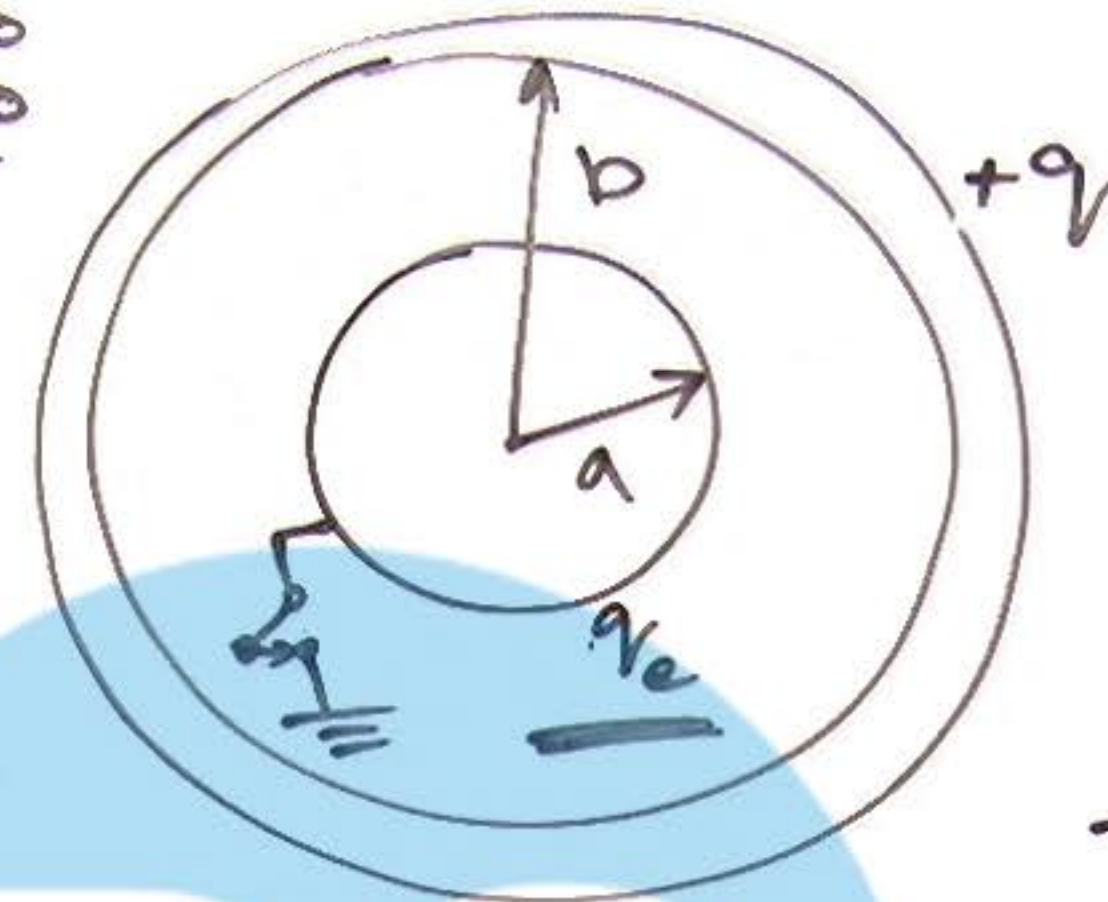
$$q \left(\frac{kQ}{y} \right) + \frac{1}{2} m v^2 = q \left(\frac{kQ}{2R^3} \left(3R^2 - \frac{R^2}{4} \right) \right)$$

$v =$ —



14 Concept of Earthing a Conductor:

This makes Conductor potential ZERO.

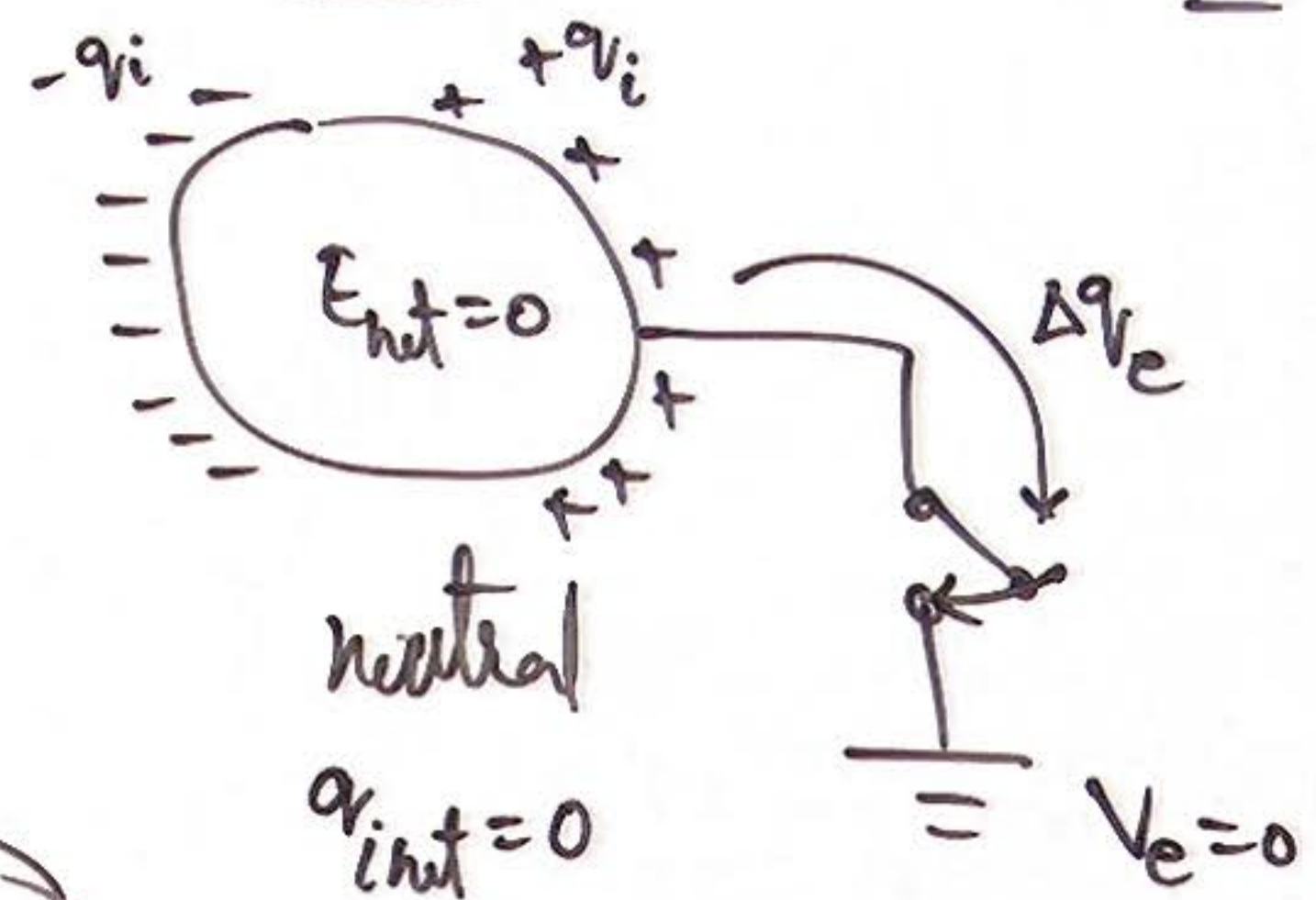
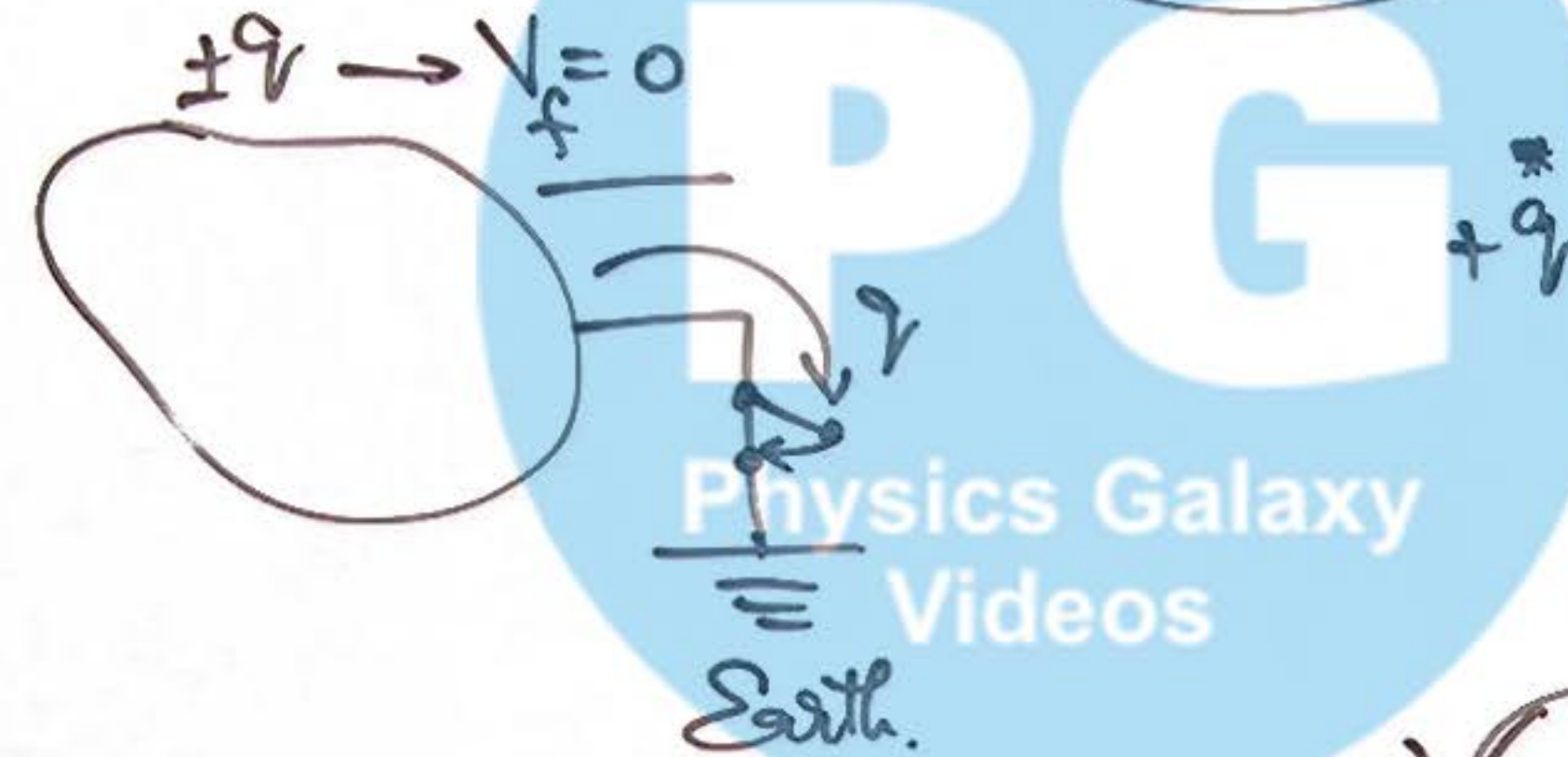


$$V_{\text{inner sphere}} = \frac{kq}{b} + \frac{kq_e}{a} = 0$$

$$q_e = -\frac{qa}{b}$$

-ve.

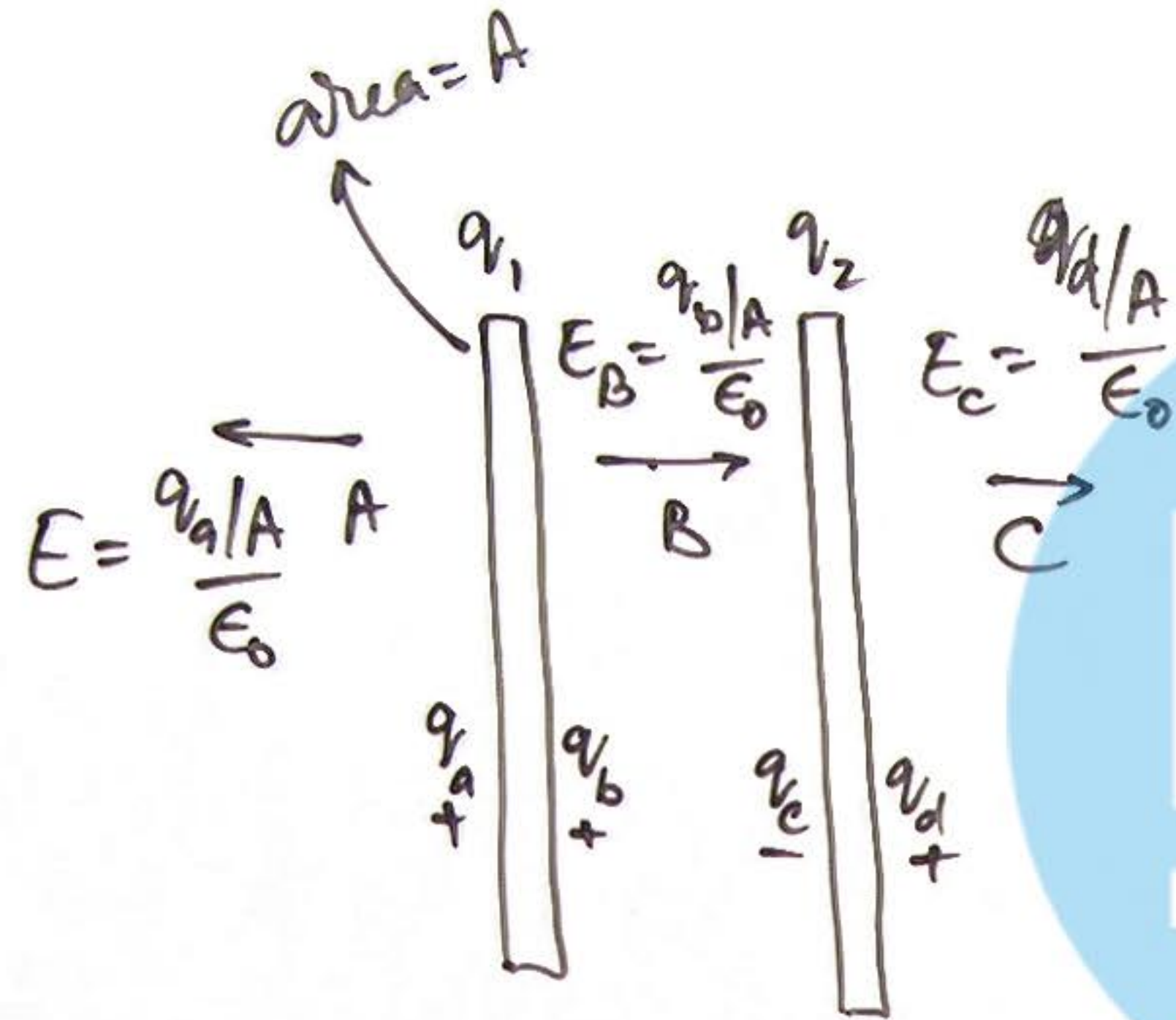
$V \rightarrow +ve.$



$V = -ve$

$\leftarrow q_{\text{net}} = -ve.$

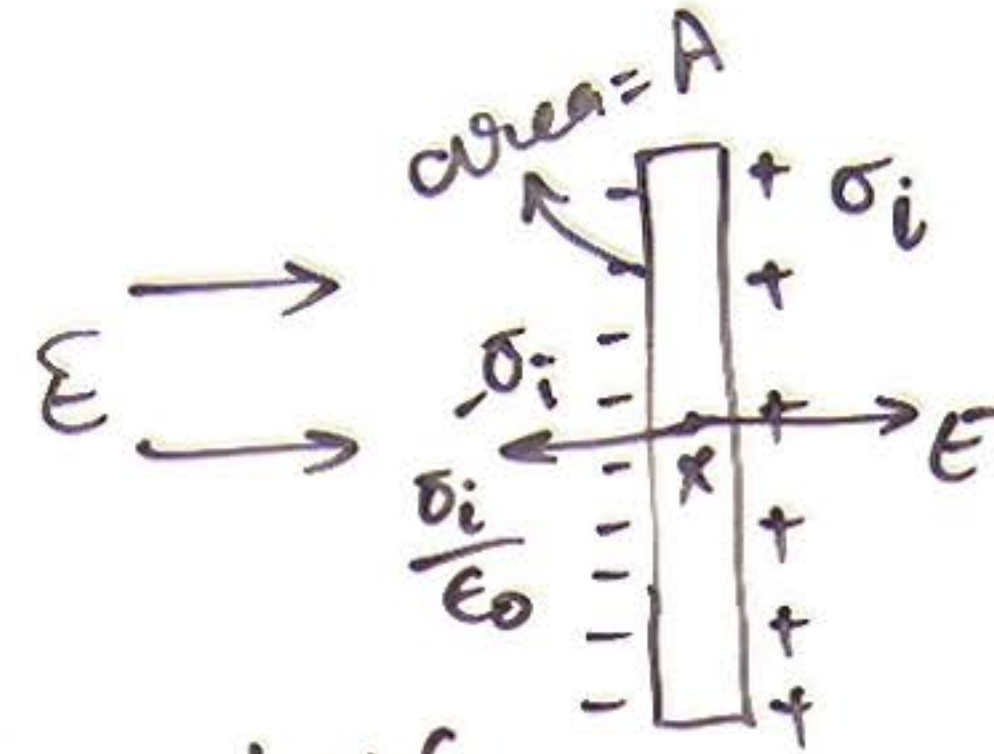
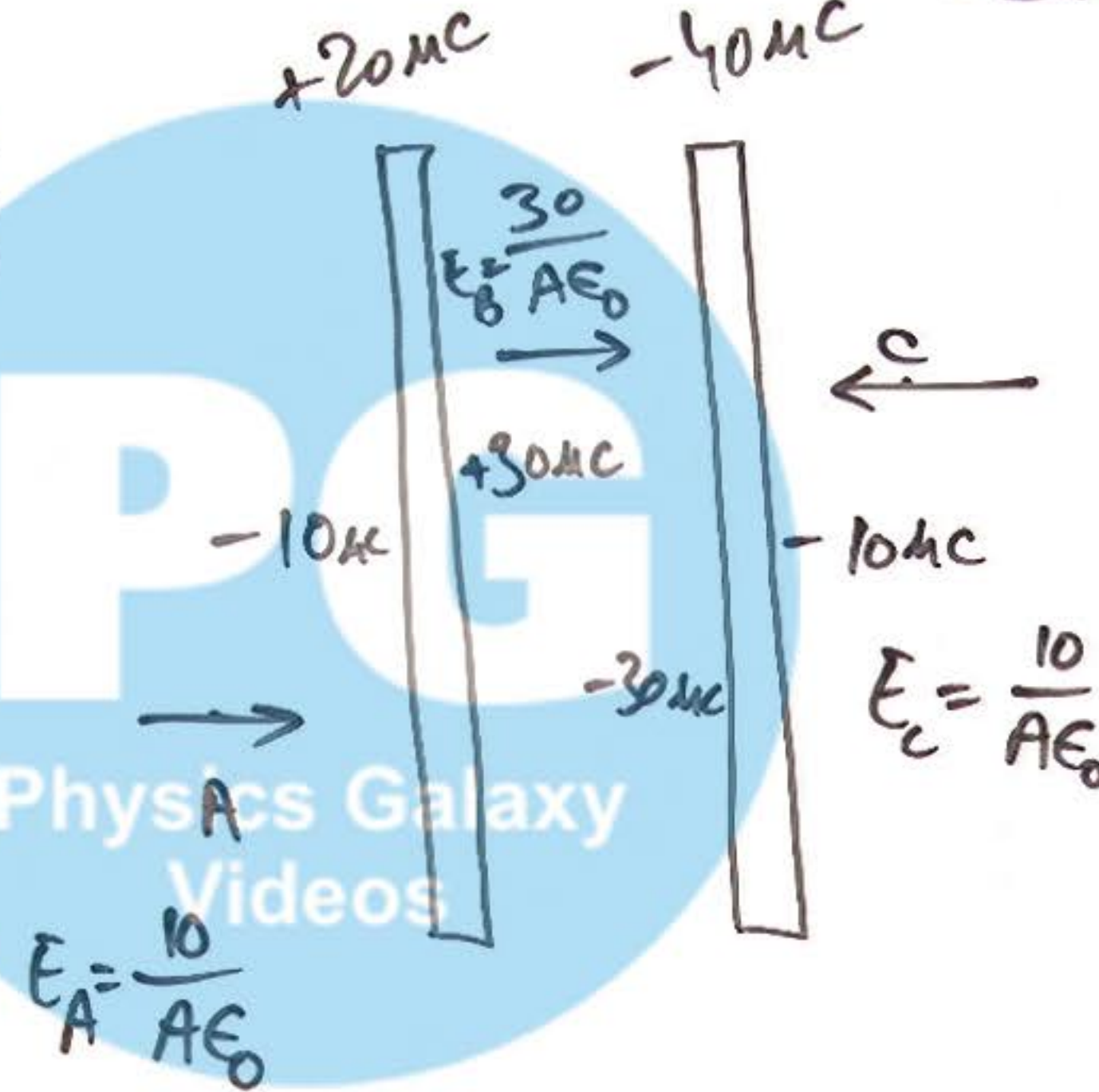
(15) A System of parallel metal plates:



$$q_c = -q_b$$

$$q_a = q_d = \frac{q_1 + q_2}{2}$$

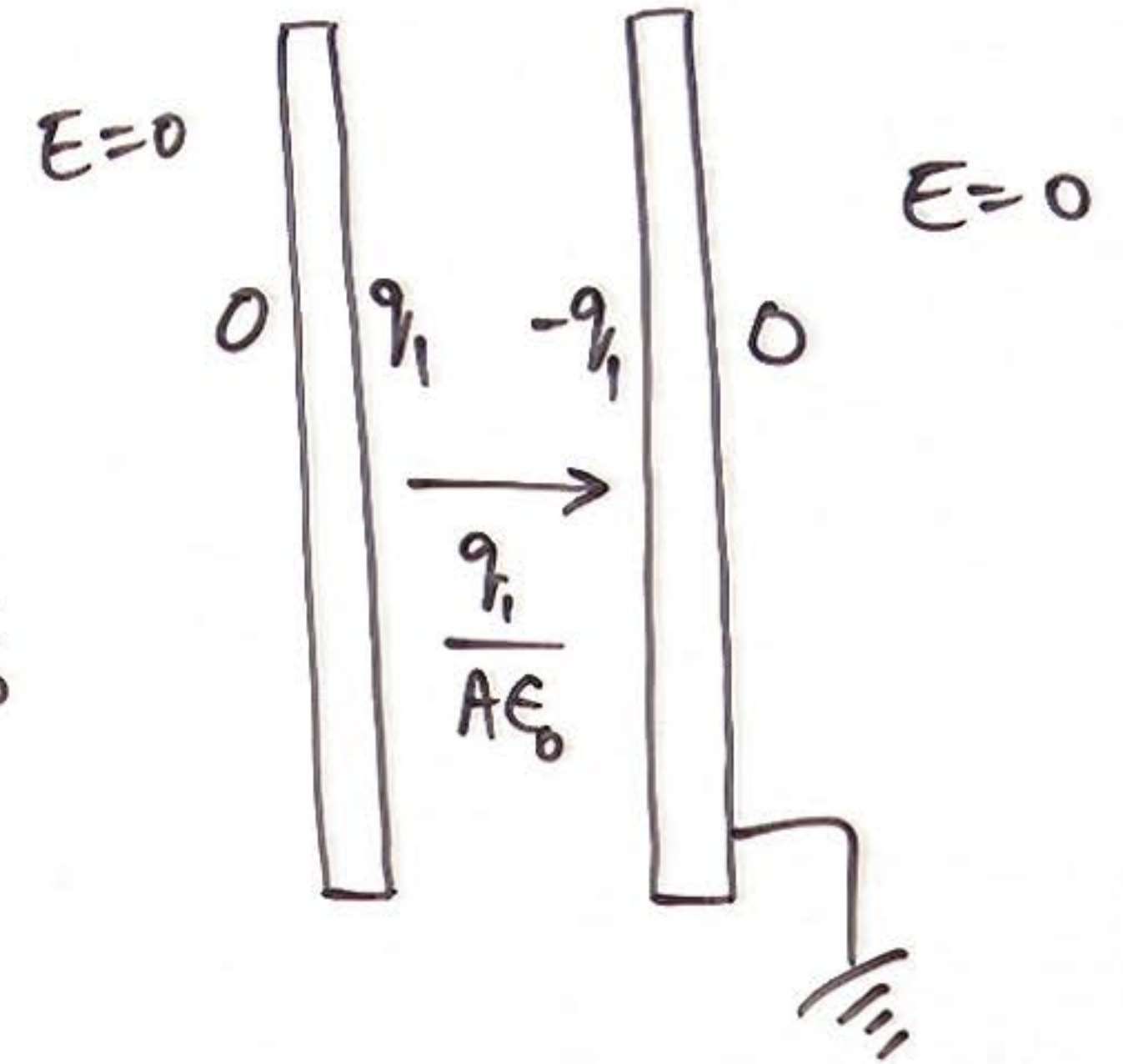
IL



$$E_{\text{net}} = 0 = E - \frac{\sigma_i}{\epsilon_0}$$

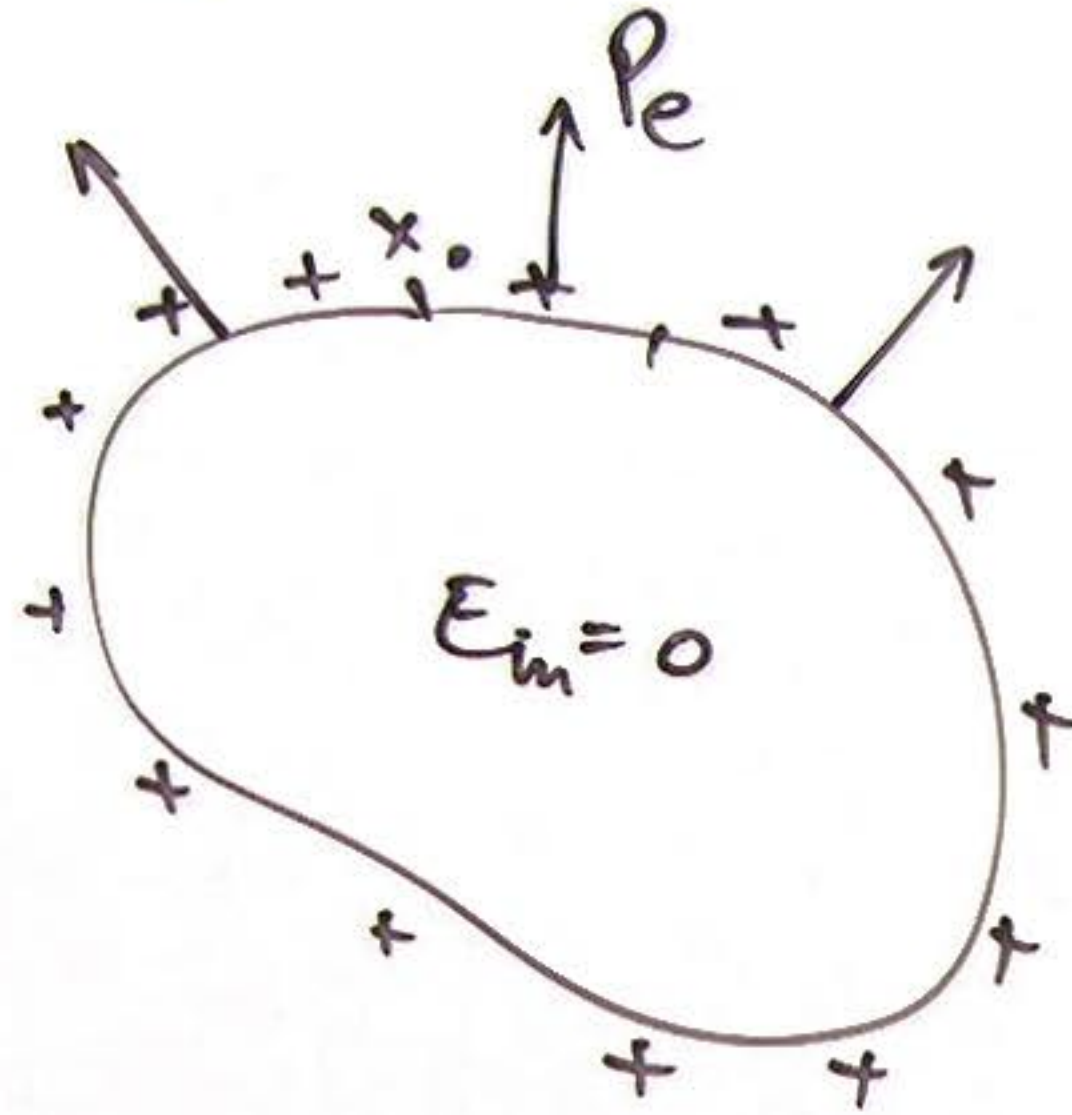
$$\Rightarrow \boxed{\sigma_i = E\epsilon_0}$$

No significant



①⑥ Electric Pressure & Energy density of EF:

due to surface charge



$$P_e = \frac{\sigma^2}{2\epsilon_0} = \frac{1}{2} \epsilon_0 E_{\text{net}}^2 = u_e$$

Energy density of EF.
(J/m³)

for all EF, ^{field} energy per unit vol.

$$E_{\text{net}} = \frac{\sigma}{\epsilon_0}$$

$$\Rightarrow \sigma = \epsilon_0 E_{\text{net}}$$

$$u_e = \frac{1}{2} \epsilon_0 E_{\text{net}}^2 \quad \text{J/m}^3$$

Self energy → W/D in charging a body.

SE of a uniformly charged metal sphere
for non metal sphere

$$U_{S_2} = \frac{3}{5} \frac{kQ^2}{R}$$

$$U_{S_1} = \frac{kQ^2}{2R}$$

$$U_S = \int \frac{1}{2} \epsilon_0 E^2 \cdot dV$$